

VIKAS SINGH

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Summary

Data Engineer with ~ 2 years of experience building and modernizing a large-scale multi-tenant healthcare data platform supporting 120+ tenants and 72 heterogeneous source systems. Hands-on ownership across S3-based ingestion, Snowflake transformation and aggregation, data distribution, application integration. Designed scalable tenant provisioning, deployment automation and data-validation frameworks. Strong production experience in Snowflake, SQL, Python, and Redshift, with additional expertise in Spark and Databricks

Experience

Hakkoda (an IBM Company)

Dec 2024 – Present

Associate Consultant – Data Engineering

- Worked end-to-end across a multi-tenant healthcare data platform spanning S3-based ingestion, Snowflake raw/silver/gold layers, distribution to serving schemas, and Spring Boot application integration.
- Contributed to migration of a Redshift-based platform to Snowflake while Redshift remained live in production, covering 25,000+ SQL queries, 8,000+ database objects, and workloads up to 5B+ rows; resolved Snowflake-vs-Redshift SQL and data-type compatibility issues.
- Worked on transformation and aggregation pipelines executing 10,000+ SQL transformations to produce 200+ Gold-layer analytical tables consumed by downstream applications.
- Optimized a large Snowflake aggregation workload from ~ 9 hours to ~ 2.5 hours (~ 72% runtime reduction) through SQL and warehouse-level tuning.
- Supported migration of the application persistence layer from Spring Boot + MySQL to Spring Boot + Snowflake, writing Snowflake code and test cases for application/database integration.
- Built a queue-triggered tenant provisioning utility (Tenant Builder) that sets up a new tenant end-to-end – including permissions and ~ 8,000 SQL objects in ~ 3 hours, across 5 environments and 3 Snowflake accounts.
- Developed a Patch Installer to propagate schema/database changes across multiple environments and tenant databases, reducing manual deployment effort and errors.
- Built tenant/schema comparison utilities to detect structural drift between tenant databases, improving platform reliability.
- Designed a Data Validator (Streamlit UI + CLI + CI/CD integration) supporting schema, row-count, row-hash, statistical, duplicate, and custom-SQL validation on datasets up to ~ 100K rows, with automated Excel-based reconciliation reporting.
- Created and executed test cases across SQL migration, tenant-level processing, and application integration; validated the platform across ~ 120 tenants and 5 environments prior to rollout.
- Enabled downstream data and ddl exports from Snowflake back to S3 for selected tenant use cases, in addition to core S3-to-Snowflake ingestion.

Projects

DataValidator Pro | Python, Streamlit, Docker

[Docker](#) | [GitHub](#)

- Built an enterprise-grade data validation platform to compare and reconcile records across different data sources (PostgreSQL, MySQL, Snowflake, Redshift, SQLite, CSV, Excel).
- Implemented validation strategies like : row-hash, row-count, schema, null-value, duplicate, statistical profiling, and custom-SQL — automating migration testing and cutting manual QA effort.
- Containerized the application with Docker, enabling single-command deployment with browser-based validation, result analysis, and downloadable audit-ready reconciliation reports.

Education

Swami Keshvanand Institute of Technology

2021 – 2025

B.Tech, Computer Science / Artificial Intelligence

CGPA: 9.7

Technical Skills

Production Experience: SQL, Snowflake, Amazon Redshift, Python, MySQL, ETL/ELT, Data Migration, Data Validation, Data Platform Engineering, Data Transformation & Aggregation, Amazon S3, Spring Boot integration

Tools & Cloud: Java, C++, Docker, Git, GitHub, Bitbucket, dbt, Streamlit, AWS CloudWatch, AWS CodePipeline, AWS CodeBuild, DynamoDB, Secrets Manager

Currently Expanding: Apache Spark, PySpark, Databricks, Delta Lake, Apache Airflow, Spark Structured Streaming

Certifications

SnowPro Core – Snowflake

Apache Spark Developer – Databricks